Julius Schneider

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EDUCATION

ETH Zurich (Bachelor in Computer Science) — **Zurich (CH)**
Focus on Computer Vision, ML and Autonomous Systems — 09/2023 – 09/2027

Purdue University (Exchange Semester) — **West Lafayette (USA)**
Joined PART (https://www.purdueaerial.com) and led field-testing and Computer Vision development — 08/2026 – 01/2027

Kantonsschule Kreuzlingen (Swiss Matura) — **Kreuzlingen (CH)**
(Swiss High School, English speaking class) — 06/2019 – 06/2023
GPA: 5.4/6; Core subjects Economics and Computer Science. Matura Paper: 6.0/6

WORK EXPERIENCE

Computer Vision Engineer CDDS (https://www.cdds.ai) — **Zurich (CH)**
Worked on Pose Estimation Models and End-to-End Learning approaches for flight controls — 02/2026 – 08/2026
Part-time Role

- Worked on expanding the Computer Vision pipeline with a pose estimation model for final trajectory predictions optimized for low-latency inference on edge hardware (NVIDIA Jetson Orin Nano)
- Engineered synthetic data generation pipelines in NVIDIA Isaac to train models for scenarios where real-world data is dangerous to collect
- Represented CDDS at EUROSATORY 2026 in Paris, discussing operational deployment requirements with defense executives and military procurement officials.
- Led cooperation with CIHBw (Germany), DARPA, DIU (USA) and Taskforce Drones (Switzerland) during field testing

Teaching Assistant ETH Zurich — **Zurich (CH)**
Teaching Assistant in the Courses Introduction to Prog. & Digital Design and Computer Architecture — 02/2025 – 06/2026
Part-time Role

- Prepared and presented weekly classes to >30 Students
- Reviewed and corrected the official final exam

Falke Digital — **Kreuzlingen (CH)**
(Drone photography and image videos of Lake Constance region, www.falkedigital.ch) — *11/2018 – Present*
Founder

- Built and operated high-performance FPV drones in cluttered environments to capture dynamic footage
- Directed and produced professional image films for customers like Mercedes-Benz Switzerland
- Achieved a substantial growth in revenue: 2022 turnover: CHF 17k; 2023 turnover: CHF 41k

PUBLICATIONS AND SELECTED PROJECTS

MIVE: Markerless Inertial Vector Extraction for Monocular Quadrotor Pose Estimation (Paper)
Authored a paper about the developed data-generation pipeline and evaluated against conventional approaches — **Zurich (CH)**

- Developed a novel pipeline to generate ground truth data for thrust vector estimation in UAVs — 02/2026 – 07/2026
- Evaluated our approach against both synthetic and ArUco based data gathering approaches

Second Place at European Defense Tech Hackathon
Built an autonomous low-cost Drone Detection System using Microphones and Cameras — **Zurich (CH)**

- Sensor activated by the sound of an incoming drone, started searching and classifying the UAV — 11/2025
- Sound detection via spectrogram analysis using a CNN and Drone classification with a finetuned YOLO model
- Worked on the software and pitched the concept to a Jury of Investors and Defense Representatives

Third Place at the ETH PhysicalAI Hackathon organized by Helbling
Built a pipeline that estimates the traveled distance of a sewer cleaning robot solely based on video data — **Zurich (CH)**

- After preprocessing we used an Optical Flow algorithm with Heuristics to judge the traveled distance — 05/2026
- Additionally trained a CNN to flag frames in which cracks or obstructions were observable in the pipes

Semester Spokesperson for ETH Computer Science
Student representative in Digital Design and Computer Architecture — **Zurich (CH)**

- Represented the interests of >500 Students in front of the board of Professors — 04/2024 – 09/2024
- Facilitated regular meetings with faculty and administration to address students' concerns

SKILLS, INTERESTS

Languages	German native, English fluent, French fluent, Russian intermediate
AI & Robotics	PyTorch, OpenCV, NVIDIA Isaac Sim, C++, C, Gazebo, PX4, Linux (Ubuntu), Git, Docker Various Projects in the areas of Drone detection, and end-to-end learning for control Theory. Matura Paper: Development of a Python-based trading algorithm which performs fully automated buying and selling decisions for stocks. Backtested against SMI and other benchmarks. Grade: 6.0/6.0
Skills	Forward Deployed Engineering, Rapid development of prototypes, Licensed UAV Pilot,